ECON 470 Homework 3.3

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Summarize the Data

Question 1

A tibble: 6 × 5

year	Mean Star Rating	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<dbl>	<int>
2010	2.97	346.99	0.0648	59579
2011	3.26	388.31	0.0844	55249
2012	3.36	410.65	0.0903	58169
2013	3.55	406.73	0.0828	64174
2014	3.81	451.11	0.0820	58636
2015	3.96	482.57	0.0819	60819

Star ratings, mean enrollments, and mean market shares tend to increase over time.

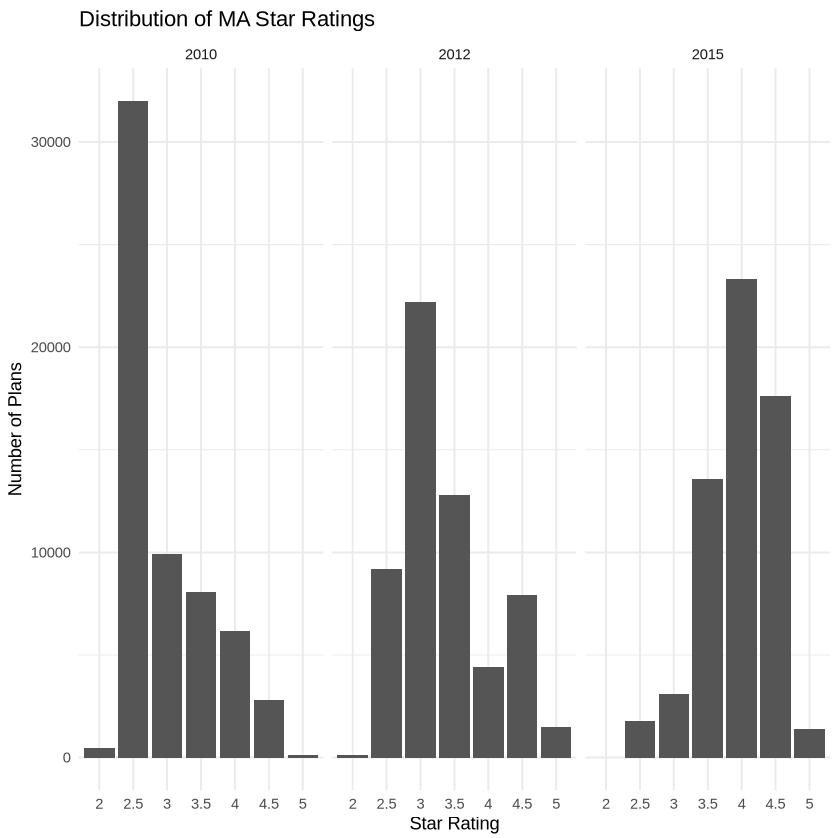
Question 2

A tibble: 6 × 4

year	Mean Enrollment	Mean Market Share	Plan Count
<dbl>	<dbl>	<dbl>	<int>
2010	107.34	0.0673	48643
2011	167.90	0.0900	12754
2012	172.73	0.0567	9077
2013	213.50	0.0600	3661
2014	257.60	0.0557	3708
2015	314.21	0.0405	4680

The plans without a star rating tend to have lower average enrollments and market share compared to plans with a star rating.

Question 3



The distribution of star ratings has shifted to the right over time, meaning plans received higher star ratings (on average) in later years. The most common star rating in 2010 was 2.5, in 2012 it was 3, and in 2015 it was 4.

Question 4

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=====================================================================
                              Dependent variable:
             -------------------------------------------------------
                                   Market Share
               2010      2011      2012      2013     2014     2015
                (1)       (2)       (3)       (4)      (5)      (6)
---------------------------------------------------------------------
3 Stars      0.013***  0.035***  0.031***  0.062*** 0.027*** 0.027***
             (0.002)   (0.002)   (0.003)   (0.003)  (0.006)  (0.005)

3.5 Stars      0.001   0.019***  0.013***  0.038*** 0.027*** 0.039***
             (0.002)   (0.003)   (0.003)   (0.003)  (0.006)  (0.004)

4 Stars       -0.004*    0.003   -0.025*** 0.017***  0.007   0.011***
             (0.002)   (0.003)   (0.004)   (0.003)  (0.006)  (0.004)

4.5+ Stars   -0.008*** -0.009*** -0.013*** 0.022***  0.007   0.023***
             (0.003)   (0.003)   (0.003)   (0.003)  (0.006)  (0.004)

Constant     0.063***  0.070***  0.079***  0.049*** 0.064*** 0.061***
             (0.001)   (0.002)   (0.003)   (0.003)  (0.006)  (0.004)

---------------------------------------------------------------------
Observations  18,993    19,260    20,895    23,755   22,793   22,949
R2             0.004     0.019     0.023     0.021    0.007    0.008
Adjusted R2    0.004     0.019     0.023     0.021    0.006    0.008
=====================================================================
Note:                                     *p<0.1; **p<0.05; ***p<0.01
```

In general, higher star ratings are associated with greater market share across years.

Estimate ATEs

Question 5

A tibble: 5 × 2

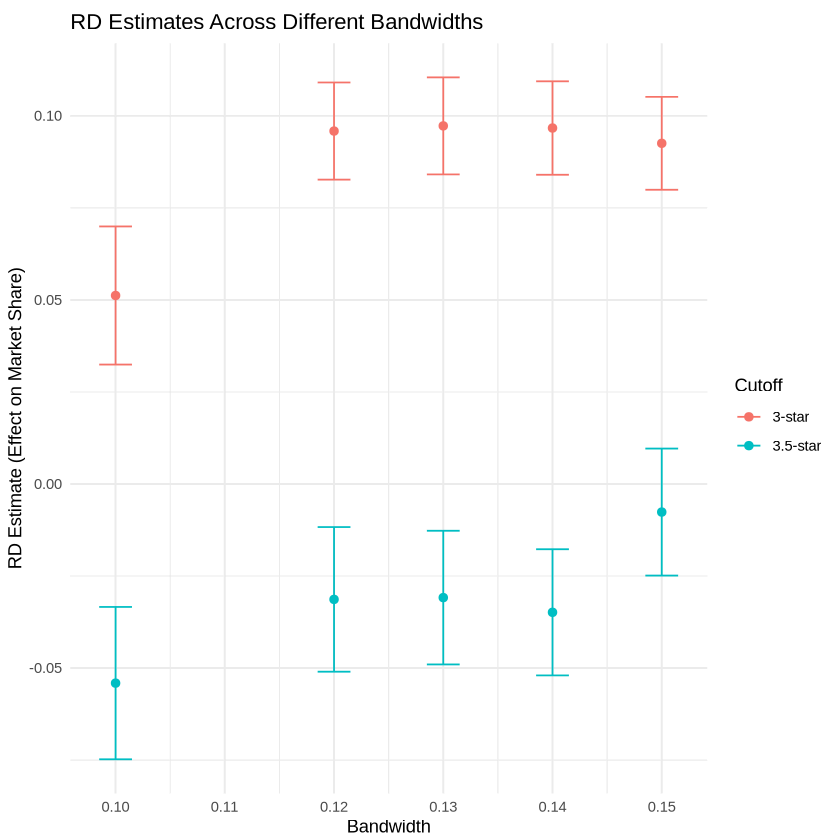
Star Rating	Plans Rounded Up
<dbl>	<int>
3.0	4425
3.5	4605
4.0	5518
4.5	1095
5.0	30

Question 6

A tibble: 6 × 3

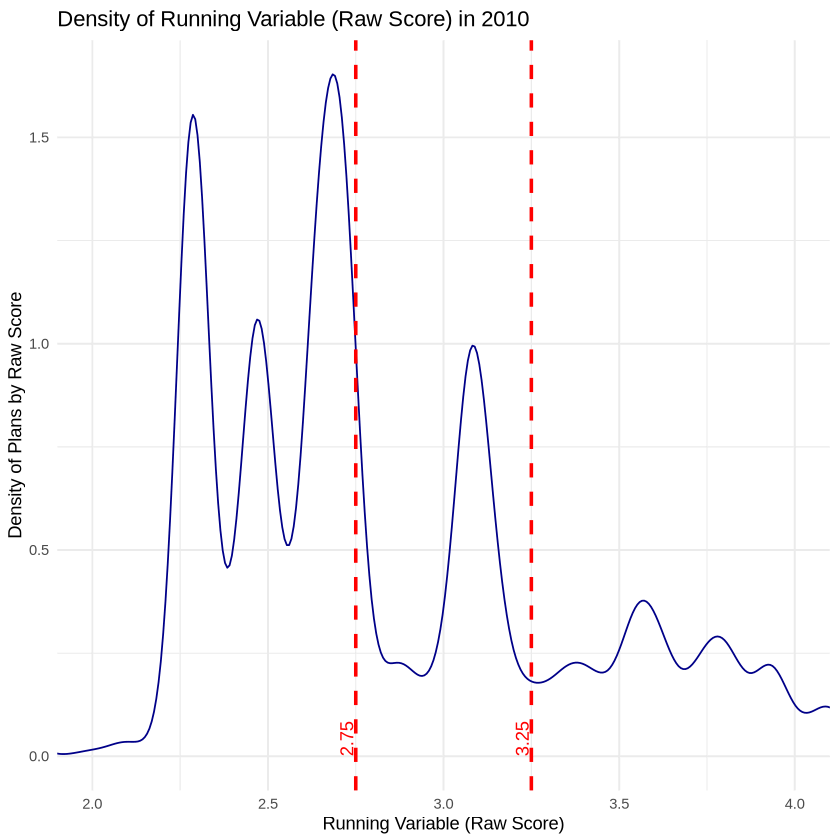
Coefficient	3 Stars	3.5 Stars
<chr>	<dbl>	<dbl>
Intercept	0.016	0.102
Rounded	0.055	-0.040
Running Score	-0.740	0.417
Rounded × Score	0.958	-0.467
Observations	5309.000	1882.000
R-squared	0.059	0.015

Question 7



My results are somewhat sensitive to the choice of bandwidth, tending to increase slightly (on average) as the bandwidth increases. The effect of receiving a 3-star rating on enrollments is around 0.05, but increases to nearly 0.10 at higher bandwidths. The effect of receiving a 3.5-star rating on market share is negative for all bandwidth values, but increases slightly as bandwidth increases.

Question 8

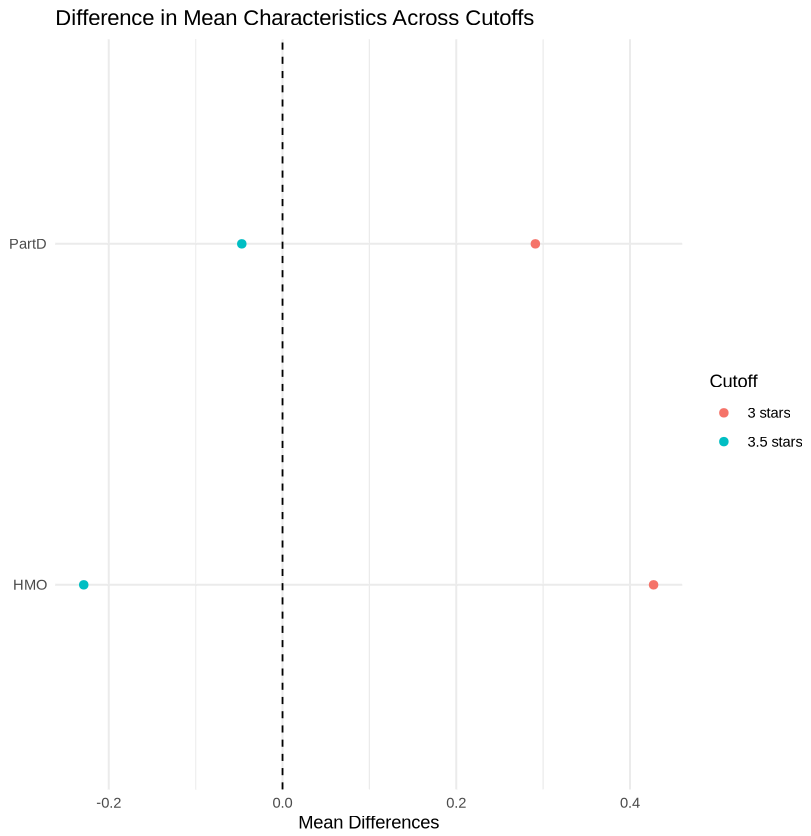


The distrubution of the running variable does not seem to change meaningfully before vs. after the 3.0 and 3.5 thresholds. There is no clear spike immediately to the right of the thresholds, so there is not strong evidence that contracts manipulate the running variable.

Question 9

A tibble: 4 × 3

Cutoff	Characteristic	Diff
<chr>	<chr>	<dbl>
3 stars	HMO	0.42716545
3 stars	PartD	0.29107216
3.5 stars	HMO	-0.22885889
3.5 stars	PartD	-0.04690552



The table and love plot indicate that around 40% more plans are HMOs at the 3-star threshold vs 2.5-stars. Also, plans just above the 3.0 threshold are more likely to have Part D. In contrast, the differences for the 3.5 star threshold are negative but smaller in magnitude. Plans just above the threshold value for 3.5 stars are slightly less likely to be HMOs or to have Part D. This suggests some differences in HMO and Part D status near the 3-star threshold and slight evidence of differences at the 3.5-star threshold.

Question 10

Based on my analysis, increasing a star rating has a slight positive effect on enrollments at the 3-star threshold and a small negative effect at the 3.5-star threshold. The RD estimate at 3 stars is around 0.05 and statistically significant, while the estimate at 3.5 stars is around -0.04 and significant with a bandwidth of 0.125. This could be because customers perceive 3 stars as significantly better than 2.5 stars, while the jump from 3 to 3.5 stars might be seen as less important. The results are somewhat similar across bandwidths, though the effect of receiving 3 stars increases slightly as bandwidth increases. There is no clear evidence of manipulation of the running variable, though plans just above the 3-star threshold are more likely to be HMOs and to offer Part D coverage than plans just below the threshold.